Graphic Era HILL UNIVERSITY
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End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.TECH, ME III SEMESTER
Name of the Paper: REFRIGERATION AND AIR CONDITIONING
Paper Code: TME 506

Time: 3 hour | Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO 1

a. What is refrigeration and how does it differ from cooling what are the various methods through which we can achieve refrigeration.

b. Explain why reversed Carnot cycle is not used in refrigeration and derive an expression for the cop of Carnot cycle.

c. An air refrigerating system used for food storage provides 25 tons of refrigeration. The temperature of air entering the compressor is 7°C an temperature at exit from the cooler is 27°C. The quantity of air circulated in the cooler is 3000 kg/hr. Both the compression and expansion follow the law $PV^{1.3}$ = CONST :

(i) COP of the cycle

(ii) Power required by the compressor per ton of refrigeration.

Q2. (2X10=20 Marks) CO 2

a. Define how subcooling and superheating effects the performance of the vapour compression cycle.

b. The temperature limits of an ammonia refrigerating system are 25°C and -10°C. If the gas is dry at end of compression, calculate the COP of the cycle assuming no undercooling of the liquid ammonia:

'Temperature (°C)	Liquid heat (kJ/kg)	Latent heat (kJ/kg)	Liquid entropy (kJ/kg-K)
25	298.9	1166.94	1.1242
−10	135.37	1297.68	0.5443

c. In a vapour compression refrigeration system using R-12, the evaporator pressure is 1.4 bar and the condenser pressure is 8 bar. The refrigerant leaves the condenser sub-cooled to 30°C. The vapour leaving the evaporator is dry and saturated. The compression process is isentropic. The amount of heat rejected in the condenser is 13.42 MJ/min. Determine:

(i) refrigerating effect in kJ/kg.
(ii) refrigerating load in TR.
(iii) C.O.P.

Q3. (2X10=20 Marks) CO 3

a. Describe the Eco-Friendly refrigerants and discuss the important features and applications of natural refrigerants.

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b. What are the harmful environmental effects of conventional refrigerants? Name any one eco-friendly refrigerant which is being commercially used and discuss its important characteristics.
c. Illustrate the properties of refrigerant that are considered before selection.

Q4. (2X10=20 Marks) CO 4

a. Derive an expression for COP of an ideal vapour absorption system
b. In an absorption type refrigerator, the heat is supplied to NH3 generator by condensing steam at 2 bar and 88 per cent dry. The temperature in the refrigerator is to be maintained at - 4°C.Find the maximum C.O.P. possible. If the refrigeration load is 18 tonnes and actual C.O.P. is 72 per cent of the maximum C.O.P., find the mass of steam required per hour. Take the temperature of the atmosphere as 25°C. (h_{fg}= 2201.6 kJ/kg)
c. 0.2 kg/s of moist air at 45°C (DBT) and 10% RH is mixed with 0.3 kg/s of moist air at 25°C and a humidity ratio of 0.018 kgw/kgda in an adiabatic mixing chamber. After mixing, the mixed air is heated to a final temperature of 40°C using a heater. Find the temperature and relative humidity of air after mixing. Find the heat transfer rate in the heater and relative humidity of air at the exit of heater. Assume the barometric pressure to be 1 atm.

Q5. (2X10=20 Marks) CO 5

a. The amount of air supplied to an air conditional hall is 300 m^3/min. The atmospheric conditions are 35°C and 55% RH. The required conditions are 20°C DBT and 60% RH. Find out sensible heat and latent heat removed from the air per minute. Also find sensible heat factor for the system.
b. Describe the principle and working of the rotary compressors and briefly explain various types of rotary compressors with the neat sketch.
c. What is the significance of the condenser in refrigerating system and describe its working principle.